

Calibration of Production Transient Alert Brokers: Preparing Decision Infrastructure for LSST

Poster talk script · Pallavi Sati · NAM

Full version 670 spoken words $\approx$ 4:45 at 140 wpm · trimmed $\approx$ 3:15 · minimal $\approx$ 2:20 (see page 2)

1 WHO I AM, AND WHAT THIS IS

Hi, I'm Pallavi Sati. I'm a student, and this is independent work — no privileged access to these classifiers or their training data, only their public outputs cross-matched with public spectroscopic labels. That constrained the study, but it also defines it: **an external audit**, of the kind anyone outside a collaboration can run.

Rubin may produce ten million alerts a night, while spectroscopy stays extremely limited. Broker probabilities already rank and filter candidates — so before relying on them for automated decisions, we should ask whether they mean what they claim. **This poster is a diagnostic of that question, not a solution to it.**

2 DATA AND GROUND TRUTH

I test that against independent spectroscopic classifications from the ZTF Bright Transient Survey — a large, systematically selected sample of bright transients. The spectroscopic class is my reference label, and neither system trained on it. The audit covers **1,576 ALeRCE objects** — ALeRCE being a production alert broker — and **278 from NEEDLE**, a specialist rare-transient classifier with three classes.

3 DOES IT CLASSIFY WELL? — THE PREMISE

First, the premise. ALeRCE classifies SN Ia and SN Ibc at about **83%**, but SN II is a clear weakness at **49.5%**. The ranking is still informative: for SN Ia versus the rest, **ROC-AUC is 0.92**. The ordering carries real information.

4 CALIBRATION

The numbers attached to that ordering are another matter. Calibration asks whether a stated confidence matches how often the classifier is right. ALeRCE's average gap is **0.26** — in one reliability bin, objects near **54%** confidence were correct about **83%** of the time. It is systematically underconfident.

5 TEMPERATURE SCALING

Temperature scaling corrects that: it rescales the probabilities with one fitted parameter, and being monotonic it never changes which class comes top. Held out, calibration error falls from **0.259 to 0.015** — repairing what the number means without altering a single prediction. At a fixed 0.8 threshold, accepted objects rise from **nine to 395**, at about 88% purity.

6 BUT THE CORRECTION IS NOT UNIFORM

It improves SN Ia and SN Ibc while making SN II worse. And NEEDLE is overconfident in aggregate, its classes needing corrections in different directions — one global fix serves neither system fully.

7 THE DECISION LAYER — WHAT THE TERMS MEAN

That brings me to the decision layer, and the terms matter here.

The setting is **selective classification**: instead of forcing a label onto everything, the system may abstain on what it is least sure of. **Coverage** is the fraction accepted automatically; **selective risk** is the error rate among only those accepted. As coverage falls, selective risk should fall too — the risk–coverage curve on the poster is that trade-off in one line.

Abstained objects are **deferred** — which means more photometry, a second opinion, or human review, not automatically a spectrum.

8 THE CHECKER

To defer, you need something to rank by. The obvious candidate is the classifier's own confidence — but that is exactly what I have shown is unreliable. So I train a **checker**: a small second model estimating the risk that ALeRCE's top prediction is wrong, from the shape of the whole fifteen-class vector.

It reaches **ROC-AUC 0.836**, against 0.689 for raw confidence and **0.752 for a control knowing only the predicted class** — that control matters, or the checker might just be relearning that SN II is hard. It clears it by **0.083**, interval excluding zero.

Defer the riskiest 20% and **47.8% of all errors** land there; random deferral would give 20%.

9 LIMITATIONS

The scope is limited: retrospective, a deliberately constructed sample of bright transients on mature light curves — not stream prevalence, and not early-alert performance. Rare classes are thin: 55 SLSN, 21 TDE.

10 WHAT'S NEXT — LEARNING TO DEFER

So the checker gives me one tested ingredient — an **error-risk score**. It ranks which predictions look least trustworthy. It does not say where a spectrum is most worth spending.

That is the gap the next stage fills: a **budget-aware deferral layer**. The idea, **learning to defer**, is that a system learns when to hand off rather than answer. Selection would weight error risk by **scientific value**, **observability** and **exposure cost**, choosing the best targets within a night's budget. **Proposed, not tested.**

11 CLOSE

So: calibration repairs the probability scale; the checker ranks which predictions appear least trustworthy; and science-aware allocation of telescope time remains to be built.

Decision-framework vocabulary — quick reference

Calibration

Whether a stated confidence matches the observed frequency of being correct. Distinct from accuracy (is the top class right?) and from discrimination (does the score rank correct above incorrect?).

Expected calibration error (ECE)

The average gap between stated confidence and observed accuracy, computed over confidence bins. Zero is perfect. ALeRCE: 0.259 raw, 0.015 after scaling.

Temperature scaling

A one-parameter post-hoc correction. Take logs of the probabilities, divide by a fitted T , renormalise with a softmax. $T < 1$ sharpens, $T > 1$ flattens. Monotonic, so the predicted class never changes.

Selective classification

Allowing the system to abstain rather than forcing a label onto every object.

Coverage

The fraction of objects accepted automatically. Its complement is the deferral rate.

Selective risk

The error rate among the accepted objects only — not over the whole sample.

Risk–coverage curve

Selective risk plotted against coverage: the whole accept/defer trade-off in one line. Lower is better.

Error capture

Of all the classifier's errors, the fraction that ends up in the deferred queue. Random deferral of 20% captures 20%; the checker captures 47.8%.

Checker (error-risk score)

A second, small model estimating the probability that the main classifier's top prediction is wrong. Validated by ranking (ROC-AUC 0.836) and calibrated in its own right (binary ECE 0.019).

Predicted-class-rate control

A baseline that scores an object using only its predicted class's historical accuracy. It tests whether the checker learns object-level difficulty or merely which classes are hard.

Deferral

Routing an object for further evidence — more photometry, a second system, human review. Not automatically spectroscopy.

Learning to defer

A framework in which a system learns when to hand off to a human or instrument instead of predicting. Proposed here, not implemented.

Budget-aware allocation

Selecting the target set that maximises expected scientific return — error risk $\times$ value $\times$ observability — subject to total exposure cost staying within the observing budget.

Timing and cuts

Full script = 670 spoken words ≈ 4:45. Per-beat counts: §1 102 · §2 57 · §3 41 · §4 46 · §5 57 · §6 32 · §7 89 · §8 110 · §9 31 · §10 81 · §11 24.

- **Trimmed, ≈ 3:15** — drop §3, §6 and §9 entirely (104 words); shorten §1 to "Hi, I'm Pallavi Sati — a student, and this is independent work using only public broker outputs and public spectroscopic labels, so this is an external audit" plus the Rubin paragraph (saves ~30); cut §7's third paragraph and §10's first (saves ~50); drop §8's middle paragraph if still long.
- **Minimal, ≈ 2:20** — keep only §1 (short form), §2, §4, §5, §8 and §11. That is the shortest version that still states the premise, the calibration result, the threshold consequence and the checker.
- **Never cut:** the 54%/83% example; the nine-to-395 figure; the 0.752 control; "proposed, not tested."

If they ask

Isn't the checker just relearning that SN II is hard?

That is what the class-only control at 0.752 is for. The checker clears it by 0.083 with a bootstrap interval excluding zero, so there is object-level signal beyond class identity. The shape-only variant does *not* clear the control — the signal lives in the full fifteen-class vector.

Is the risk score itself calibrated, or only well-ordered?

Both. Out-of-fold, its own binary ECE is 0.019 and Brier score 0.144.

Why not just lower the threshold instead of calibrating?

You cannot choose a defensible threshold without an audit like this one, and raw scales are not comparable across systems — ALerCE fitting $T = 0.36$ while NEEDLE fits 1.55 is exactly that problem.

How much of the deferral gain is real?

Modest but genuine: 47.8% of errors captured against 20% for random deferral, with a perfect oracle capped near 69% at that coverage — so just over halfway from random to ideal.

Does calibration make the classifier more accurate?

No. Temperature scaling is monotonic — not one predicted class changes. It repairs what the number means, which changes *which objects clear a fixed gate*, not whether they were classified correctly.

Why is NEEDLE in a poster about brokers?

NEEDLE is not a broker — it is a specialist rare-transient classifier. It is included as a contrasting deployed system: it fails in the opposite direction to ALerCE, which is what shows that a single global correction cannot be assumed to work.

Every figure quoted is reproducible from [results/tables/](#) in github.com/pallavi-2000/transient-calibration-audit, branch **alerce-selective-deferral**.